



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



ROBOTIC ARM

Ignite Session

by
MaRS Rover Students Club

Motivation

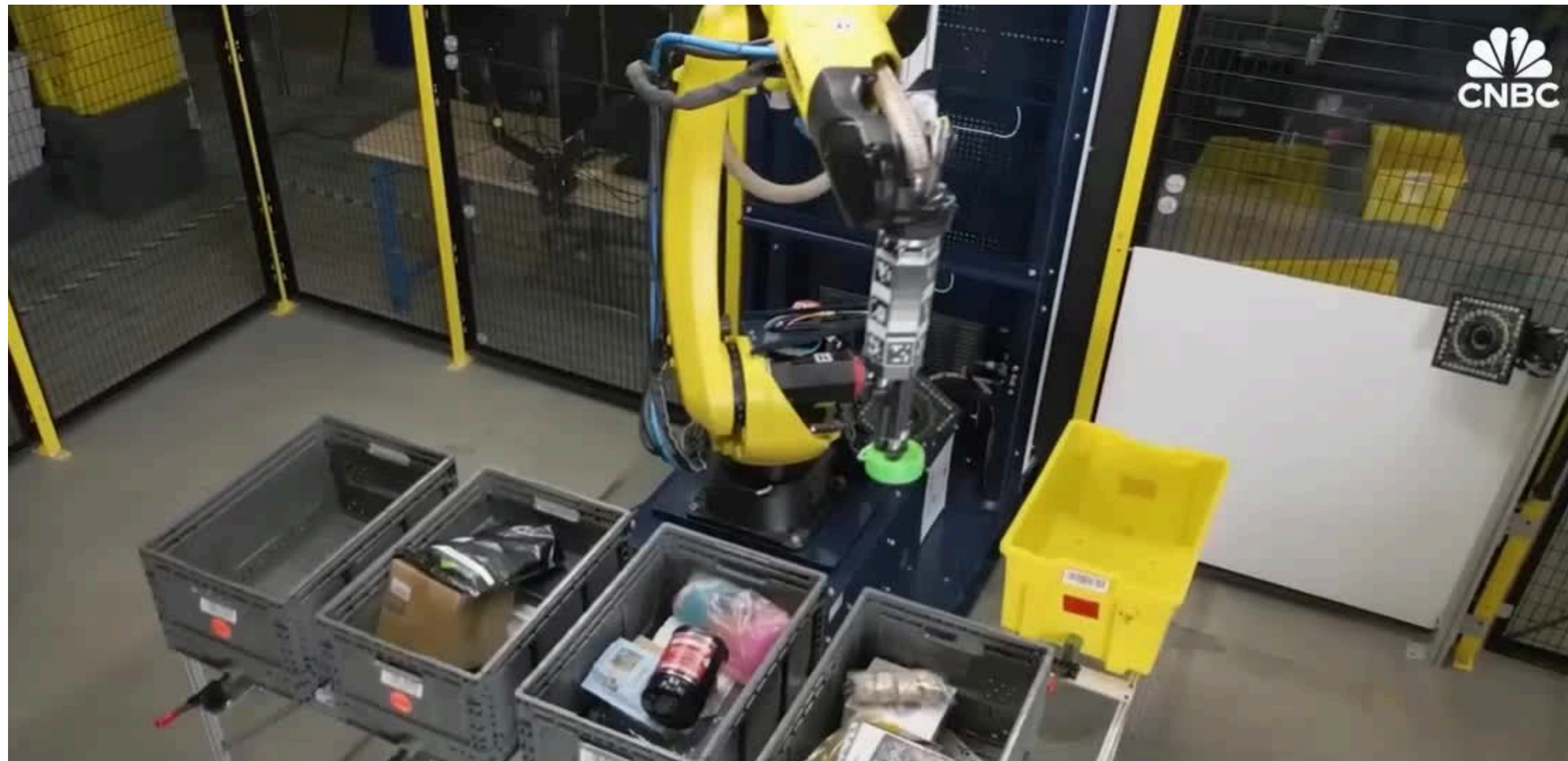


“What do Doc Ock from Spider-Man, the Winter Soldier, and General Grievous from Star Wars have in common?”



Motivation

“Have you ever wondered how Amazon sorts and packages millions of products every day?”



Prerequisites

#1

Enthusiasm for robotics

#2

Love for experimenting, even if things sometimes catch fire (in simulations, of course).
- Trial and error is part of the game.

#3

Willingness to break, fix, and break it again.
- Manipulators aren't built perfectly the first time

Anatomy of a manipulator



Anatomy of manipulator

- The entire robotic manipulator is like your whole arm, from your shoulder down to your fingertips
- It performs precise tasks by moving and positioning tools or objects in its workspace.
- Actuators: Muscles of the robot—motors, hydraulics, or pneumatics that move joints.
- End Effector: The “hand” or tool—grippers, welding torches, spray nozzles, or scientific drill samplers.”

Kinematics

How do you calculate the position of the end effector in space? We use forward and inverse kinematics to model “where” and “how to get there.”

Dynamics

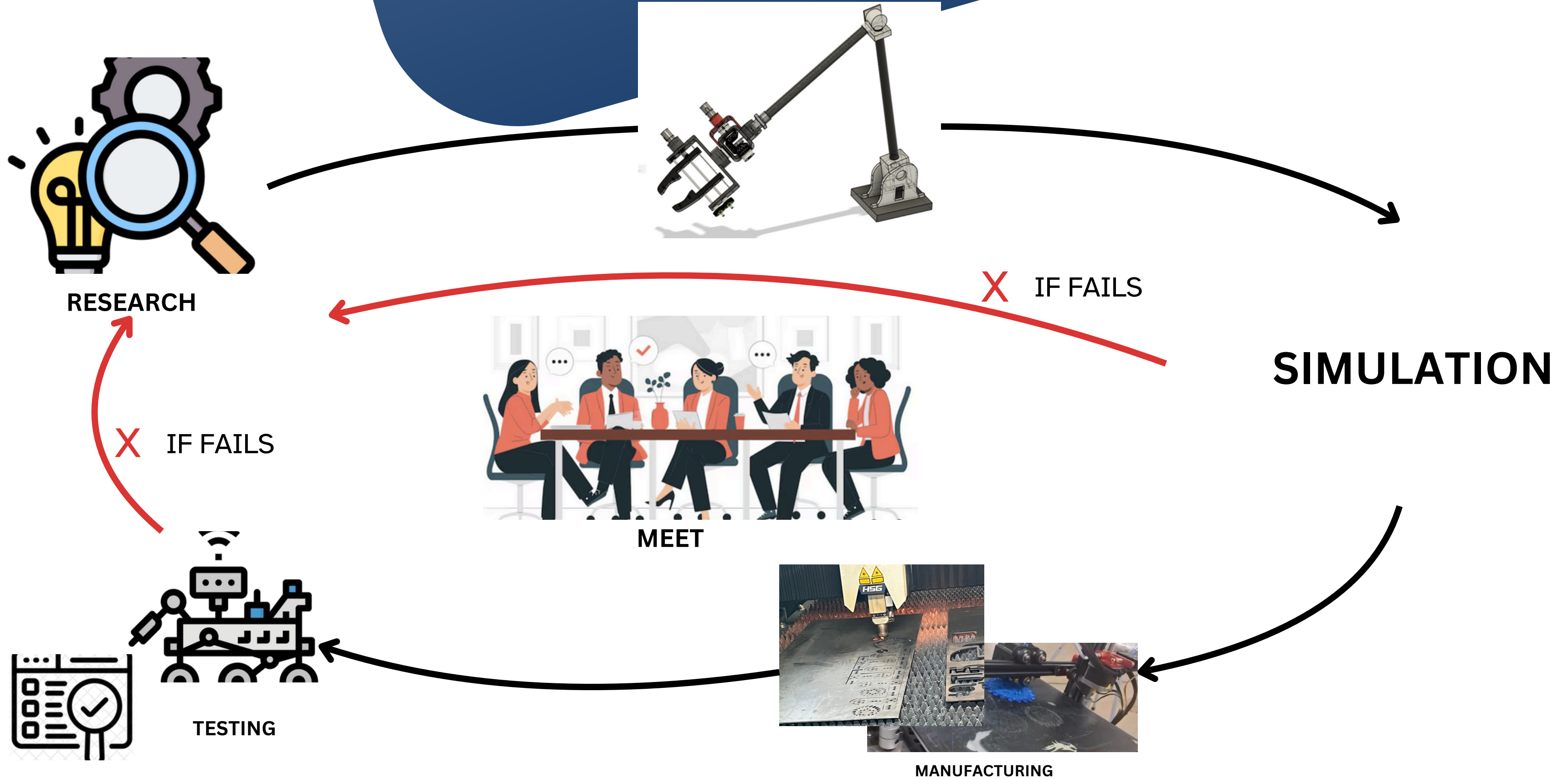
Once we know “where,” how much torque or force do our motors need? We model mass, inertia, gravity, and friction.

Control Systems

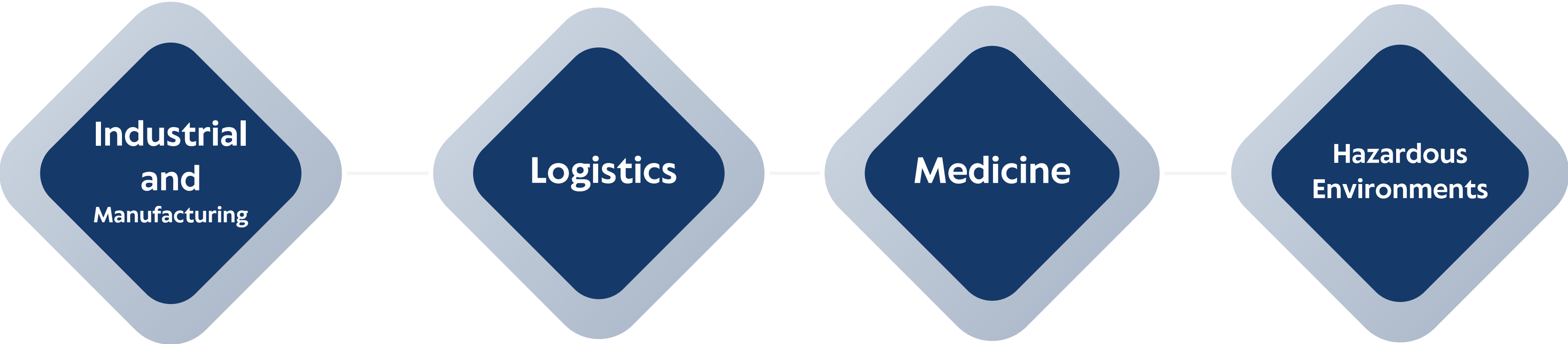
PID controllers, trajectory planning, and real-time feedback loops let the arm follow precise paths smoothly.”

**THE SCIENCE BEHIND
THE MOTION**

IDEA TO IMPACT



REAL-TIME Applications



**Industrial
and
Manufacturing**

Spot welding, painting car
bodies, inserting circuit boards

Logistics

Sorting parcels, palletizing
goods, autonomous mobile
robots (AMRs) with arms

Medicine

Da Vinci surgical robot,
rehabilitation exoskeletons,
prosthetic limbs

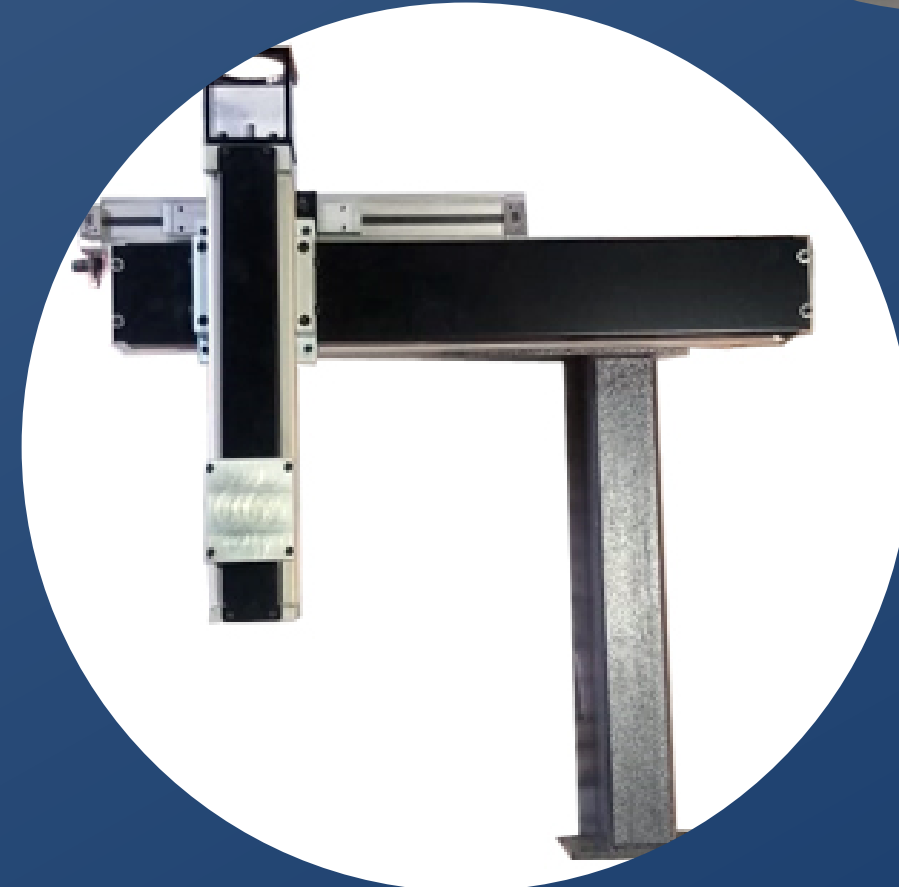
**Hazardous
Environments**

Bomb disposal robots, nuclear plant
maintenance, deep-sea sampling
And Farming ...and many more...

The Technical Aspect of Robotic Arms

- ◆ **Soft Robotics:** Flexible arms made of silicone or fabric.using compliant Mechanisms that bend and flex rather than rotate using flexible elements — no hinges or bearings needed.— safe for humans and delicate tasks.
- ◆ **Lightweight Materials:** Use of carbon fiber and 3D-printed composites — stronger, faster, and energy-efficient.
- ◆ **Modular joints & Tendon-Driven Actuation:** Cable-based joints like human fingers — compact and precise.
- ◆ **Harmonic & Worm Gears:** High precision and self-locking — perfect for space and energy-limited environments.”

Various Industrial Manipulators



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THANK YOU

